

Self-Driving Car · Premium Course

3-session series (2 hrs each) · 11-15 yrs · from KES 9,000

Premium course outline **PREMIUM**

Overview

Build a chassis-based car that navigates on its own, finishing with a race day.

DETAIL

Track	Arduino Robotics & Electronics
Format	3-session series (2 hrs each)
Ages	11-15 yrs
Level	Intermediate → Advanced
Price	from KES 9,000
Maker Kit	Optional, KES 8,500 one-time: Arduino Uno R3 board + USB cable, Breadboard + 40 jumper wires, LEDs (red, yellow, green, white, RGB) + resistors, Push buttons, potentiometer and piezo buzzer, Light sensor (LDR) and sound sensor, HC-SR04 ultrasonic sensor + bracket, IR sensor, DHT11 temperature & humidity sensor, PIR motion sensor, SG90 servo motor, Relay module, 16×2 LCD screen (I2C), Battery holder, Storage box. Bought once and reused in every Arduino course. Kits are lent during sessions; buy one to keep building at home.
Parts pack	Optional, KES 2,500: 2WD car chassis with 2 DC motors and wheels, L298N motor driver, Second IR sensor, AA battery pack
Standalone	No
Prerequisites	Servo motor control (free)

Programme

- 1. Motorised Chassis.** Assemble the chassis and wire DC motors to an L298N motor driver, then program forward, reverse and turn controls and drive a test course.
- 2. Line Follower.** Calibrate IR sensors on black vs. white, then program the car to follow a track on its own and tune its speed on the curves.

3. **Obstacle Avoidance.** Mount an HC-SR04 ultrasonic sensor so the car detects and steers around objects, then race the finished cars on race day!

What they take home

 Working self-driving car kit

Why parents love it

“The technology behind Waymo and Tesla — built from scratch by your 12-year-old.”

How to book

Places are limited and run on scheduled dates. To book or ask a question, email **tinkerwithanne@gmail.com** with the course name (**Self-Driving Car**), your child's age, and your preferred dates. Prices start from the amount above and may vary with group size and materials.